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MILITARY SPECIFICATION

PUMPING ASSEMBLY, FLAMMABLE LIQUID,
BULK TRANSFER PMU-27/M

1. SCOPE

1.1 This specification covers one type of bulk transfer, flammable liquid pumping assembly, designated PMU-27/M.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-601	Aluminum-Alloy Sand Castings

Military

MIL-V-173	Varnish, Moisture- and Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-H-4441	Hose, Rubber, Gasoline, Lightweight
MIL-T-5021	Tests, Aircraft and Missile Welding Operators, Qualifications
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-M-8127	Meter, Volumetric Positive Displacement, Liquid, Aircraft Fuel, 60 GPM, Type MA-1
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for
MIL-C-10387	Coupling, Clamp, Pipe, With Bolts and Synthetic Rubber Gasket, for Grooved Pipe and Tube

FSC 4320

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MIL-P-10388	Pipe Fittings, One or More Ends Grooved
MIL-E-11275	Engine, Gasoline, Industrial Type, General Specification for
MIL-N-26978	Nozzle, Fuel and Oil Servicing, Over-the-Wing, Shockproof, Type MD-1 and MD-2
MIL-C-27487	Coupling Halves, Quick Disconnect-Cam-Locking Type
MIL-H-27508	Hose Assembly, Rubber GRK-6/E32R-1
MIL-R-38489	Reels, Static Discharge, Grounding RMU-17/E and RMU-18/E
MIL-H-52112	Heating Torches, Equipment Winterization, Instant Lighting
MIL-F-52308	Filter Element, Fluid Pressure

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-756	Reliability Prediction
MIL-STD-808	Finishes, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MS27022	Coupling Half, Quick Disconnect, Cam-Locking Type, Male, External Pipe Thread, Type III
MS27028	Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IV
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20325.)

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United States of America Standards Institute

S1.4	General Purpose Sound Level Meters
S1.6	Preferred Frequencies for Acoustical Measurements
Z24.10	Octave-Band Filter Set for the Analysis of Noise and Other Sounds

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. The specification makes provisions for preproduction testing.

3.2 Components. The pumping assembly shall consist of the following major components. Each component shall include all parts, controls, instruments, and accessories necessary for its effective operation.

Item or System Name	See Requirement
Pump	3.7.1
Engine	3.7.2
Filter-separator	3.7.3
Meter	3.7.4
Trailer chassis	3.7.5

3.2.1 Arrangement of components. The components of the pumping assembly shall be assembled on the trailer chassis essentially as shown on figure 1.

3.2.1.1 Location. The components shall be positioned on the trailer chassis to provide uniform distribution of weight and use of the minimum lengths of piping and tubing.

3.2.1.2 Mounting. Components that need not be removed or repositioned for use shall be securely mounted to the trailer chassis or to other major components in a manner that will insure against damage or unnecessary movement during operation and transport. If necessary, additional subbases or other fixtures required for safe and secure component mounting shall be provided.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment. When materials are used in the construction of the pumping assembly that are subject to deterioration when exposed to climatic and

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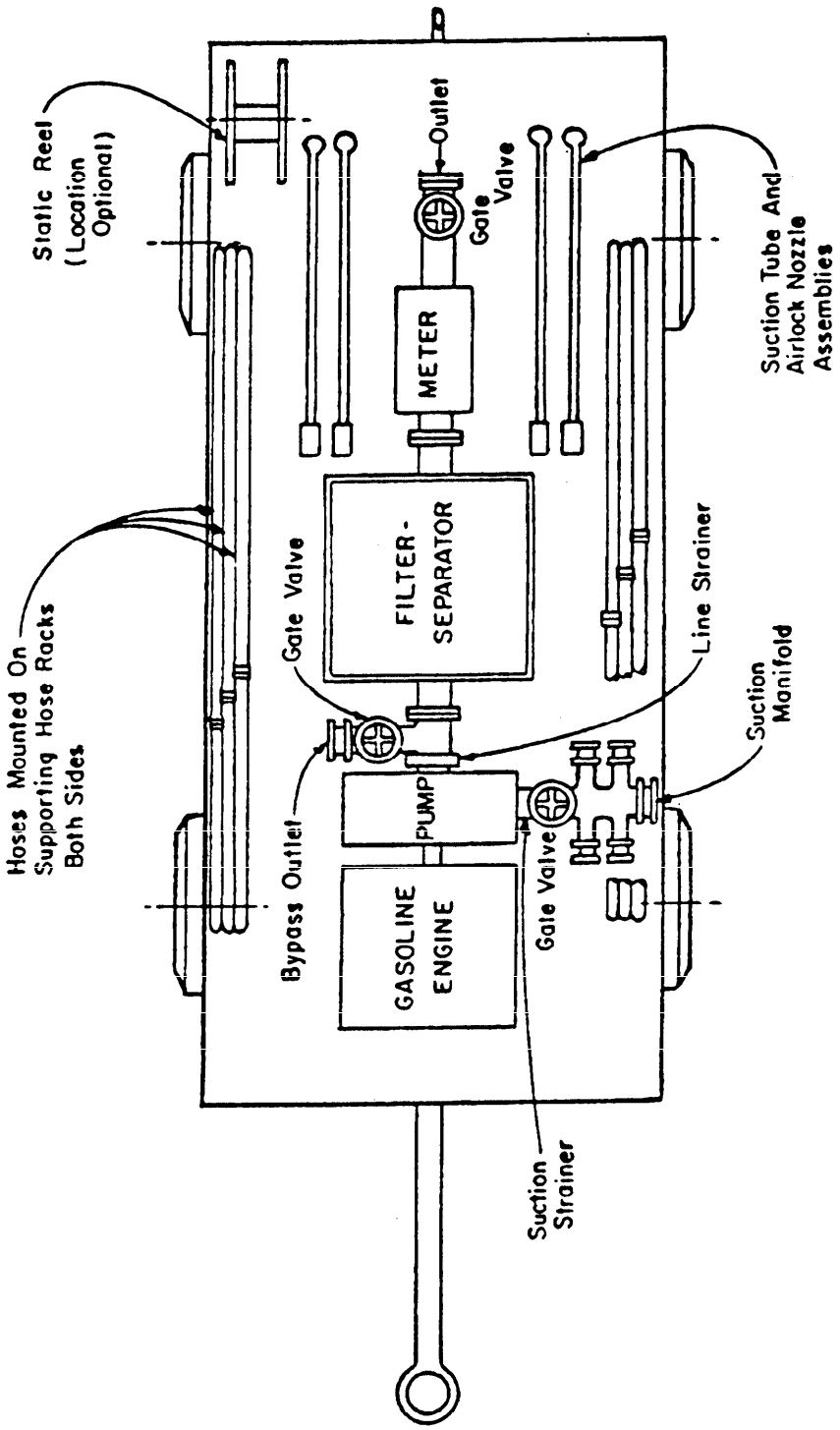


FIGURE 1. Arrangement of Components

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environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with varnish conforming to MIL-V-173. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.3 Elastomers. Elastomers used in the pumping assembly shall be suitable for the application. Elastomers that come into contact with the fuel system shall be compatible with fuels conforming to MIL-G-5572 and MIL-T-5624. All elastomers shall be satisfactory for all environmental conditions specified herein.

3.4.4 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, additives, salt spray, or atmospheric conditions likely to be met in storage or normal service. All metals that come into contact with the fuel shall be nonferrous.

3.4.4.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.5 Design and construction. The pumping assembly shall be designed and constructed to withstand the strains, shocks, vibrations, and other detrimental conditions incident to operation, maintenance, shipping, and storage with minimum loss of time for maintenance, repair, and periodic servicing.

3.5.1 Reliability. The pumping assembly shall have a predicted mean-time-between-failure (average operating time between failures) of not less than 1,200 hours. (See 6.3.1.)

3.5.1.1 The prediction shall be based upon the principles of MIL-STD-756 on the assumption that the random failure rate of a system can be estimated as a function of component random failure histories and estimates. Failure rate data and estimates used in the prediction shall be modified to account for component use times and for stresses, loads, and environments expected compared to those of the data.

3.5.1.2 The intent of the prediction shall not be to require extensive part testing to develop component failure rates. The intent of the prediction shall be to apply each component based on engineering evaluation of failure frequency data and observations of identical or similar components and designs in related environments and applications. The contractor shall have available documented failure rate data and estimates used as a basis for the prediction.

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3.5.2 Maintainability. The pumping assembly shall be designed and constructed in accordance with the design guidelines of MIL-STD-470, paragraph entitled, Establish Maintainability Design Criteria, except rather than being merely considered, the guidelines shall be applied wherever appropriate and wherever they do not conflict with other requirements specified herein. Where practicable, the design and construction of the pumping assembly shall permit maintenance and servicing under adverse weather extremes. All maintenance, servicing, and adjustments which may be economically practicable at lower than depot level shall be capable of being performed using only common handtools and equipment commonly expected in shops where maintenance can be accomplished on similar equipment.

3.5.3 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.4 The pumping assembly shall be a complete, self-contained unit assembled within the trailer framework and shall be designed for the following:

a. Defueling four 55-gallon storage drums concurrently

b. Pumping from an external source and servicing an aircraft over-the-wing

c. Defueling aircraft auxiliary fuel tanks

d. Manual starting

e. Continuous and intermittent operations

f. Pumping from an external source bypassing the filter-separator.

3.5.5 Tubing, piping, and electrical wiring. Tubing, piping, and electrical wiring shall be located in protected positions, securely fastened to frame or body structures, and provided with metal protective looms or grommets at each point where they pass through structural members, except where a through-the-frame connector is provided. Electrical wiring in open locations shall be inclosed in conduit.

3.5.6 Gate valves. Three gate valves shall be provided on the pumping assembly. A 2-inch nominal gate valve shall be installed on the suction line, a 1-1/2-inch nominal gate valve shall be provided on the outlet of the meter, and a 1-1/2-inch nominal gate valve shall be installed on the outlet side of the tee located between the outlet of the pump and inlet of the filter-separator. All valves shall be provided with fittings conforming to MIL-P-10388, except for the valves at the tee and the outlet connection of 3.5.9.2. The outlet of the valve at the tee shall be equipped with a 1-1/2-inch MS27022-9 coupling half and an MS27028-9 dust cap, types III and IX, class 1 conforming to MIL-C-27487.

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3.5.7 Static grounding reel. One RMU-18/E static grounding reel conforming to MIL-R-38489 shall be provided. Prior to mounting the reel on the trailer chassis, the paint on the contacting surfaces shall be scraped bare to decrease electrical resistance. The resistance between the grounding clamp and pumping assembly shall not exceed 10 ohms.

* 3.5.8 Instrument panel. An instrument and control panel or panels shall be provided on the pumping assembly. If more than one panel is used, all panels shall be visible from, and operable by, one operator standing in one location. All gages, switches, and other instruments shall be centrally located on the panels. The panels shall be fabricated from rigid material resistant to mechanical vibration damage. Instruments shall be grouped according to the system to which they pertain, i.e., engine instruments and controls shall be in one group, and pump instruments in another group. Panels shall be designed to permit accessibility of instruments from the rear for maintenance purposes. The following instruments and controls shall be provided and installed on the panels:

a. Choke assembly

b. Throttle assembly

c. Engine stopswitch

d. Differential pressure gage

e. Pump discharge pressure gage.

3.5.8.1 Standard engine controls. The throttle, choke, and engine stopswitch shall be of conventional design normally used throughout the automotive industry.

3.5.8.2 Pressure gage. A pump pressure gage having a dial diameter of not less than 4-1/2 inches shall be provided. It shall be a flush-mounted, dial-type gage with a minimum accuracy of at least ±2 percent of the full-scale range. The gage shall incorporate a Bourdon tube. The scale range shall be selected on the basis of the normal operating pump pressures. The normal operating pressure shall indicate at approximately 60 percent of the full-scale range. The graduations of the dial shall have 5-psi figure intervals with intermediate intervals of 1 psi. (See 6.4.)

3.5.9 Pumping assembly inlet and outlet connections

3.5.9.1 Inlet connection. The pumping assembly shall be provided with a 2-inch ID nominal suction pipe manifold approximately 12 inches in length. The manifold shall be fabricated from aluminum-alloy material suitable for the application. The manifold shall have four class 1, 1-1/4-inch male coupling halves, and one 2-inch, class 1 male coupling half conforming to MIL-C-27487. The couplings shall be so spaced that interference will not be encountered when hose sections are attached. Each coupling shall be provided with four MS27028-7 and one MS27028-11 dust caps capable of sealing and preventing loss of pump prime when attached. The caps shall be secured to the

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manifold with chains to prevent their being lost when not in use. The manifold shall connect to a 2-inch gate valve, which shall connect to a suction strainer, connected to the inlet side of the pump. The suction strainer shall have a 20-mesh screen and shall be so designed that the screen can be easily removed for inspection and cleaning.

3.5.9.2 Outlet connection. The outlet connection on the gate valve at the meter shall be provided with an MS27022-9, 1-1/2-inch, type III, class 1 male coupling half and an MS27028-9, type IX, class 1 dust cap conforming to MIL-C-27487.

3.5.9.3 Hoses, couplings, and attachments

3.5.9.3.1 Hoses. Unless otherwise specified (see 6.2), the pumping assembly shall be provided with the following hoses:

a. One 20-foot section of 2-inch suction hose conforming to MIL-H-4441

b. Four 20-foot sections of 1-1/4-inch suction hose conforming to MIL-H-4441

c. One 30-foot section of 1-1/2-inch discharge hose conforming to MIL-H-27508.

* 3.5.9.3.1.1 Coupling halves. Each suction and discharge hose section shall be provided with one male and one female coupling half on the ends of each hose. The coupling halves shall conform to MIL-C-27487. The material shall be aluminum alloy conforming to QQ-A-601.

3.5.9.3.1.2 Hose clamps. Each coupling half hose shank shall be bonded to the hose with two corrosion-resistant steel bands having corrosion-resistant steel buckles. The width of the bands shall be not less than 1/2 inch.

3.5.9.3.1.3 Nozzle. A Type MD-1, 1-1/2-inch nozzle conforming to MIL-N-26978 shall be furnished for use with the discharge hose. The nozzle shall be furnished with a female coupling half conforming to MIL-C-27487. The nozzle shall be tightly stowed to seal its inside by being supported by a dustproof socket at the spout end and a 1-1/2-inch, class 1 dust plug conforming to MIL-C-27487 at the other end, both socket and plug mounted on the frame.

3.5.9.4 Pumping system piping. The pumping system piping shall be fabricated from aluminum-alloy material of the grade and temper suitable for the application. All components installed in the piping system shall be interconnected with pipe fittings and couplings conforming to MIL-P-10388 and MIL-C-10387, respectively. The piping system shall be designed to withstand a hydrostatic pressure of 125 psi without leakage for 15 minutes.

3.5.9.5 Airlock nozzle and suction tube. Four airlock nozzle and suction tube assemblies shall be provided with the pumping assembly to facilitate pumping fuel from 55-gallon-capacity storage drums. The nozzle assembly shall be provided with a 1-1/4-inch female coupling half conforming to MIL-C-27487. The suction tube shall

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be attached to the airlock nozzle which shall be designed to be inserted into a 55-gallon-capacity fuel drum. The airlock nozzle and suction tube shall be designed to facilitate the de-drumming of a 55-gallon fuel drum and to automatically shut off the air supply to the suction hose when the drum is emptied. The airlock nozzle and suction tube assemblies shall be tightly stowed by being supported by four sockets at the tube ends and at the coupling ends by four 1-1/4-inch, class 1 dust plugs conforming to MIL-C-27487 which shall be mounted to the frame.

3.5.9.6 Line strainer assembly. A 40-mesh line strainer assembly shall be provided and located on the discharge outlet of the pump upstream from the bypass valve. The strainer shall be so designed that the screen can be easily removed for inspection and cleaning.

3.5.9.7 Filter-separator bypass piping. A 1-1/2-inch tee with a 1-1/2-inch valve shall be installed between the outlet of the pump assembly and the inlet to the filter-separator. This bypass line shall have the capability of bypassing the filter-separator and meter assembly when pumping highly contaminated products.

3.5.10 Mobility. The pumping assembly shall be designed for type II, group C mobility in accordance with MIL-M-8090 (see 6.2) with the exceptions and additions specified:

a. 40 mph over level paved highway

b. 25 mph over graded gravel road

c. 15 mph over Belgian block course

d. 15 mph over cross-country terrain.

3.5.11 Air transportability. The pumping assembly shall meet the air transportability requirements of MIL-A-8421 for transportation in the Type C-123 aircraft.

3.5.12 Welding. All welders employed in the fabrication of the pumping assembly shall be certified in accordance with class A, groups I and IV, of MIL-T-5021.

3.6 Performance

3.6.1 Environmental conditions

3.6.1.1 Operating temperatures. The pumping assembly shall be capable of performing as specified herein in ambient temperatures between -65° and +130°F. The pumping assembly shall be started within 10 minutes and operated at -20°F without engine preheat. At -65°F, it shall be started within 30 minutes and satisfactorily operated.

3.6.1.2 Storage temperature. The pumping assembly shall not be damaged by storage in ambient temperatures from -80° to +160°F for not less than 72 hours.

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3.6.1.3 Humidity. The pumping assembly shall not be damaged by operation or storage in any relative humidity up to and including 100 percent, including conditions wherein condensation takes place in the form of water and frost.

3.6.1.4 Fungus. The pumping assembly shall not be damaged by exposure to moist fungus growth such as encountered in tropical and subtropical climates.

3.6.1.5 Salt atmosphere. The pumping assembly shall not be damaged by operation or storage in an atmosphere containing salt-laden moisture such as encountered near bodies of salt water and in transportation on shipboard.

3.6.1.6 Dust. The pumping assembly shall not be damaged by operation or storage in an atmosphere containing airborne sand and dust particles such as encountered in normal and desert operations.

3.7 Details of components

3.7.1 Pump. The pump shall be of the self-priming type (rotary or centrifugal) having adequate capacity to pump grade JP-5 fuel conforming to MIL-T-5624 at a delivery rate of 50 gpm through the piping system and dispensing hose when the discharge nozzle is elevated 20 feet above the pump with the inlet of the 2-inch suction hose and fuel level located at a point 15 feet below the centerline of the pump.

3.7.1.1 Housing. The pump housing shall be constructed of cast aluminum of a grade, type, and temper to suit the application.

3.7.1.2 The pump shall be so designed that it can be removed from the pumping system. Groove-type fittings conforming to MIL-P-10388 shall be provided on the pump inlet and outlet connections.

3.7.1.3 Bearings and seals. The pump shall have long-life bearings and seals that will insure long service life with a minimum of maintenance. The pump shall be so constructed and positioned that fuel will not enter the engine oil system should a pump seal fail.

3.7.1.4 Drain plug. A drain plug having a minimum size of 1/2 inch NPT shall be provided at the lowest practical point on the pump.

3.7.1.5 Pump drive. The pump shall be directly driven by the pump engine. The pump shall be provided with a removable drive coupling to permit the pump to be easily disconnected from the engine driveshaft assembly.

3.7.1.6 Pump operation

* 3.7.1.6.1 Pump priming. The pump shall be so designed that it will prime a dry suction hose with grade JP-5 fuel conforming to MIL-T-5624 within 1 minute when the inlet of a 20-foot section of suction hose and the fuel level in the storage tank are 15 feet below the centerline of the pump.

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3.7.1.6.2 Closed discharge. The pump shall be designed to operate at rated speed against a totally closed discharge without excessive wear or damage after 5 minutes of closed discharge operation at an ambient of 70°F.

3.7.1.6.3 Dry operation. The pump shall be designed to operate without damage at maximum speed for 1 hour without liquid in the pump.

3.7.1.6.4 Pumping life. The pump shall operate satisfactorily for 125 hours with normal servicing and sparkplug change.

3.7.2 Engine. A 4-cycle, air-cooled, gasoline engine conforming to MIL-E-11275 shall be provided. The engine shall have a net reserve continuous brake horsepower rating of not less than 25 percent greater than the horsepower required for operation of the pump at 50 gpm at rated rpm.

3.7.2.1 Engine starter. An automatic recoil-type, hand-operated engine starter shall be provided. The starter shall be so designed that the engine can be started with a rope should the automatic starter fail.

3.7.2.2 Fuel tank. A fuel tank shall be mounted below the engine for the storage of gasoline for the engine assembly. It shall have sufficient capacity to provide fuel for 5 continuous hours of operation of the engine without refilling of the tank. It shall be provided with a filler neck having a minimum inside diameter of 2-1/2 inches. The fuel filler cap shall be painted insignia red, film designation BG as specified in MIL-STD-808.

3.7.2.3 Flame arrestor and exhaust pipe. An exhaust pipe shall be provided and so located that it will cause neither deterioration of the exterior pumping assembly nor detrimental health conditions in the way of noxious fumes to operating personnel. A flame arrestor shall be provided on the engine exhaust pipe assembly.

3.7.3 Filter-separator. A filter-separator rated at 50 gpm shall be provided on the pumping assembly that meets the performance requirements of MIL-F-8901.

* 3.7.3.1 An exception to MIL-F-8901 is specified in 4.6.8.1.

3.7.3.2 Housing. The filter-separator housing shall be fabricated from aluminum alloy material of the grade and temper to suit the application. The housing shall be designed to withstand a pressure of 125 psi. The housing shall hold the filter elements in a horizontal position. The housing shall have a separate water sump area. The housing shall be in the pumping system so the sump extends below the bed of the trailer assembly. The housing shall be designed to accommodate three elements conforming to MIL-F-52308 and a 200-mesh tetrafluoroethylene separator screen. The housing shall be equipped with a removable head and division plate to permit the

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removal and replacement of the filter elements. The housing shall be provided with both inlet and outlet gage connections. Each gage connection shall be clearly marked INLET or OUTLET, as applicable.

3.7.3.3 Inlet and outlet connections. The inlet and outlet connections of the filter-separator shall be 2-inch nominal pipe size and shall be located on the filter-separator in a manner to permit ease of assembly and disassembly of the filter-separator for element changes. The connections shall conform to MIL-P-10388.

3.7.3.4 Automatic water drain valve. An automatic water drain valve shall be incorporated into the filter-separator to permit the removal of all water arrested by the filter-separator. The draining capacity of the valve shall be in excess of 2 gpm at internal pressures from 20 to 75 psi. The valve shall be installed in the sump.

3.7.3.5 Water slug and control valve. The filter-separator shall be fitted with a water slug and control valve installed on its outlet side. The valve shall limit the flow through the filter-separator at 50 gpm. The valve shall automatically stop fuel discharge from the filter-separator when the incoming fuel is contaminated with a quantity of water exceeding the draining capacity of the automatic drain valve. It shall be designed to permit normal operation of the filter-separator automatically, by draining the water from the sump, should the valve close due to a water slug.

3.7.3.6 Water and fuel drain valve. A 3/4-inch gate valve shall be installed in the sump of the filter assembly to permit the draining of fuel and water from the filter-separator. The valve shall be made from aluminum alloy material suitable for the application.

3.7.3.7 Filter elements. The filter-separator shall be provided with three filter elements conforming to MIL-F-52308.

3.7.3.8 Separator screen. The filter-separator shall be provided with a 200-mesh monel screen coated with tetrafluoroethylene. The screen shall be installed on the interior of the filter-separator on the outlet piping. The area of the screen shall be not less than 300 square inches.

3.7.3.9 Differential pressure gage. A 4-1/2-inch differential pressure gage shall be provided to indicate the differential pressure across the filter-separator. It shall be designed to withstand static pressures of 125 psi. It shall be provided with a straight scale from 0 to 100 psi with the dial graduated in 10-psi figure intervals and 1-psi subdivisions. It shall be provided with overpressure stops and shall have an accuracy within 1 percent of full scale range. It shall be provided with the marking DIFFERENTIAL PRESSURE across the face of the dial. The gage shall be provided with two independently mounted Bourdon tubes made from seasoned phosphor bronze. The dial shall be coated steel with a black background and white graduations and figures. The gage shall be provided with 1/4-inch NPT external threaded back or bottom connectors and clearly marked to identify both inlet and outlet ports.

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3.7.3.10 Nameplate. The filter-separator shall be provided with a nameplate as shown on figure 2. The plate shall be fabricated of sheet aluminum not less than 0.050 inch thick.

3.7.4 Meter. One Type MA-1 meter assembly conforming to MIL-M-8127 shall be installed on the pumping assembly. The inlet connection shall conform to MIL-P-10388. The outlet connection shall be flanged.

3.7.5 Trailer chassis. The trailer chassis shall be designed to meet the requirements of MIL-M-8090 as specified for type II, group C, mobility. The design exception and additions are as follows:

a. The assembly shall be provided with 6.00 x 9, 6-ply tires and tubes.

b. Neither service brakes nor service brake tests shall be required.

c. No bumpers, fenders, or mudflaps are required.

d. Power light system shall not be required.

e. The assembly shall be provided with springs and shock absorbers designed to meet the mobility requirements specified herein.

3.7.6 Accessory storage racks

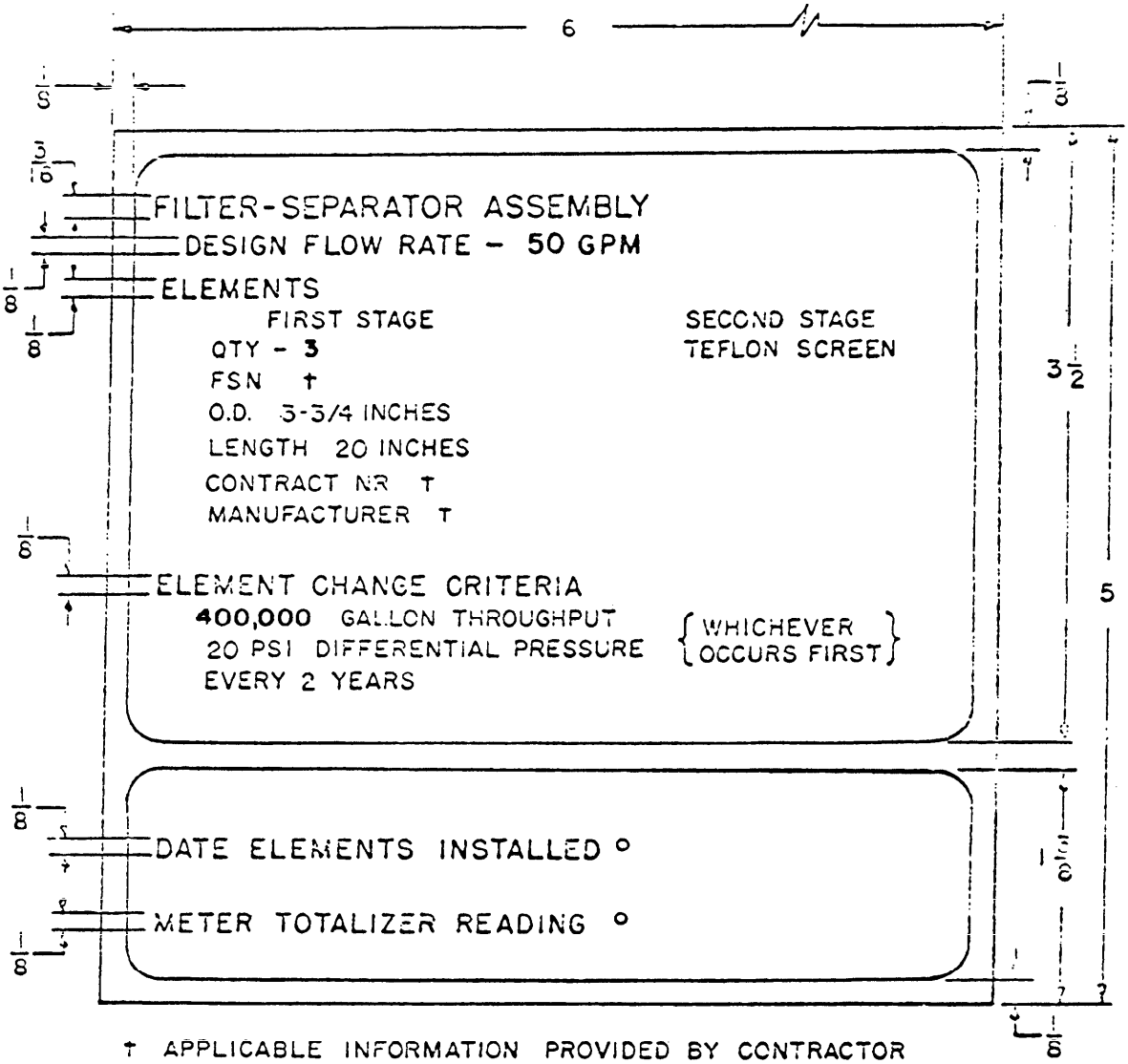
* 3.7.6.1 Hose racks. A fixed hose rack or racks shall be provided for storage of all hoses during transit in a manner to support the hoses and to permit the male end to be inserted into the female end of the same or any other hose section to preclude the use of dust covers and caps. The racks shall be provided with locks or spring buckles to prevent the hoses from unwinding during transit.

3.7.6.2 Suction tube and fuel servicing nozzle storage rack. A storage rack shall be provided to permit storing the fuel servicing nozzle, airlock nozzle, and suction tubes. The storage rack shall be attached to the pumping assembly and shall be so designed that the accessories will not fall off the trailer while in transit.

3.7.7 Winterization kit. Unless otherwise specified (see 6.2), a winterization kit shall be provided only on the preproduction pumping assembly. The kit shall include all components and materials such as weather seals, ducts, a blower, heat exchanger, et cetera, as required to meet the performance requirements for the pumping assembly. A heater conforming to MIL-H-52112 shall be furnished in the kit. The kit shall be so designed that it can be installed and removed with a minimum of maintenance and installation time. The kit shall be designed to produce sufficient heat to start and operate the pumping assembly within 30 minutes when ambient temperatures are at -65°F.

3.7.8 Handbook and record compartment. A compartment shall be provided on the pumping assembly for containing the equipment operator's handbooks and the equipment

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† APPLICABLE INFORMATION PROVIDED BY CONTRACTOR

° PROVIDE APPLICABLE SURFACE TO ALLOW DATA TO BE MARKED BY GOVERNMENT PERSONNEL

NOTES:

1. ALL LETTERING SHALL BE 3/32-INCH HIGH, UNLESS OTHERWISE NOTED
2. BACKGROUND SHALL BE BLUE
3. DIMENSIONS IN INCHES, TOLERANCES ± 1/32

FIGURE 2. Nameplate

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maintenance and inspection record. The compartment shall be of sheet metal and weatherproof construction and shall be a minimum size of 9 by 11-1/2 by 2 inches.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Acoustic noise levels. The acoustic noise levels produced by the pumping assembly shall not exceed the values specified in table I when measured as specified in 4.6.6.

TABLE I. Maximum Noise Level Limits

Octave Band Center Frequencies (Hz)	Sound Pressure (Decibels db Ref 0.0002 dynes/CM^2) Overall Sound Pressure Level
63	100
125	100
250	100
500	95
1,000	92
2,000	88
4,000	88
8,000	85

3.10 Electromagnetic interference. The pumping assembly shall comply with the requirements specified in MIL-STD-826, method 7001.

3.11 Screw threads. Screw threads shall conform to Handbook H28.

3.12 Dimensions. The pumping assembly shall not exceed the following maximum dimensions:

		Inches
a.	Height	35
b.	Width	45
c.	Length	100

3.13 Finishes and protective coatings

3.13.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. The exposed surfaces shall be finished in yellow, color code 13538.

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3.14 Operational markings

3.14.1 Control marking. All valves, gages, controls, and indicators used in the operation of the pumping assembly shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not fade or cause them to become eradicated. Tags shall not be used.

3.14.1.1 All gages, switches, and other instruments provided on the instrument panel shall be clearly identified as to their functions.

3.14.2 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed on or near the instrument panel. The instructions shall be clear, concise, and adequate to enable operation of the pumping assembly without damage to the equipment or injury to personnel. Instruction panels shall be fabricated of sheet aluminum or sheet zinc of not less than 0.050 inch thickness, anodized or etched to produce raised markings with a black or other dark color background, with a border of not less than 1/4 inch.

3.14.3 Transportation data plate. A transportation data plate conforming to MIL-P-514 shall be securely attached to the pumping assembly as near as practical to the nameplate. The transportation data plate shall contain the following information:

SHIPPING WEIGHT_____POUNDS

LOADING CUBAGE_____CUBIC FEET

CENTER-OF-GRAVITY LOCATION:

VERTICAL_____FROM_____

HORIZONTAL_____FROM_____

OVERALL HEIGHT_____

OVERALL WIDTH_____

3.14.4 Lifting instructions plate. An instruction plate containing all information necessary for transportation personnel to rig a safe lifting sling from common cable or spreader bar components and to safely lift the pumping assembly with a 3-g acceleration shall be securely attached to the outside of the housing assembly near the transportation data plate. The lifting instruction plate shall be of the same material and prepared in the same manner as the transportation data plate.

3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16 Workmanship. The pumping assembly, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular

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attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.16.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the pumping assembly or any of its components they shall be held or limited accordingly.

3.16.2 Cleaning. The pumping assembly shall be thoroughly cleaned of loose, spattered, or excess solder; welding slugs; stray bolts and nuts; pipe compound; rust metal particles; and other foreign matter during and after final assembly. All weld beads shall be thoroughly cleaned and weld spattering shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the pumping assembly shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Apparatus. Insofar as is practicable, apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to insure laboratory accuracy.

4.3.2 Test fuel. Unless otherwise specified (see 6.2), test fuel to be pumped shall be grade JP-5 fuel conforming to MIL-T-5624. For fuel for the pump engine, see 4.6.9.1.2.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One pumping assembly shall be subjected to the tests specified in 4.4.3.

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4.4.2 Test report. After the preproduction tests are completed, three completed copies of a test report in accordance with MIL-STD-831 shall be supplied to the procuring activity. The test report shall include test results on all components covered by this specification and applicable specifications referenced herein. If preproduction or qualification tests are not available from the component supplier, it shall be the responsibility of the prime contractor to conduct tests and supply such data as necessary to assure that components meet the applicable specification requirements.

* 4.4.2.1 The test report shall contain the evaluations referred to in 4.6.9.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling plan and tests . . . . . See 4.5.2.

4.5.1 Individual tests. Each pumping assembly shall be subjected to the following tests:

a. Examination of product . . . . . See 4.6.1

b. Mechanical operation . . . . . See 4.6.2

c. Pressure test . . . . . See 4.6.3.

4.5.2 Sampling plan and tests. One pumping assembly shall be selected at random from every fifty, or fraction thereof, produced during each 30 calendar days of the production period, and subjected to the following tests:

a. Parking brake test . . . . . See 4.6.4

b. Suction lift and flow demonstration . . . . . See 4.6.5.1.

4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, items still on hand or produced later shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of pumping assemblies on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

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4.5.2.3 Defects in pumping assemblies already accepted. The investigation of a test failure could indicate that defects may exist in pumping assemblies already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The pumping assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Mechanical operation. A thorough check shall be made of the pumping assembly to insure proper functioning of all mechanical parts, accessibility of all controls and levers, interchangeability of removable parts, and other checks that will insure every degree of safety and performance.

4.6.3 Pressure test. The pumping and piping systems of the pumping assembly shall be subjected to a hydrostatic pressure of 125 psi for 15 minutes. There shall be no leaks in the system as a result of this test.

4.6.4 Parking brake test. The parking brake test shall be conducted as specified in MIL-M-8090.

4.6.5 Pump demonstration

* 4.6.5.1 Suction lift and flow demonstration. The 2-inch, 20-foot section of suction hose shall be attached to the pumping assembly with the inlet inserted into an underground fuel storage tank having a fuel level 15 feet below the centerline of the pump. The section of 1-1/2-inch discharge hose, 30 feet in length, shall be elevated 20 feet with the nozzle attached. Additional piping or hoses shall be provided to form a recirculating system. After the pump is initially primed, the pump shall circulate 10,000 gallons of fuel through the unit at a rate of 50 gpm. The pump engine shall be shut off and the suction hose disconnected from the pump suction stub and reconnected after the fuel has been drained from the hose. The engine shall be restarted and the time recorded for the pump to prime. A total of 10 disconnects, engine starts, and priming demonstrations shall be performed. The pump shall prime itself satisfactorily within 1 minute without extra priming aids.

4.6.5.2 De-drumming demonstration. One end of each of the four 1-1/4-inch suction hoses shall be attached to each of the suction stubs in the pumping assembly. The airlock nozzle and suction tube shall be attached to the other end of each hose. Each of the four suction tubes shall be inserted into a 55-gallon drum filled with fuel. After the pump is initially primed, the pumping assembly shall be used to partially fill five additional empty 55-gallon capacity fuel storage drums. After each drum is

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emptied, each suction tube shall be alternately removed and inserted into each of the partially filled drums and the previously emptied drums refilled. This cycle shall continue until a total of one hundred drums are emptied and refilled. The pump shall retain its prime while the drums are emptied.

4.6.5.3 Pumping life. The pumping assembly shall be installed in a circulating system as specified in 4.6.5.1 and subjected to a 125-hour life test as follows:

a. Pumping for 25 hours at a rate of 10 gpm

b. Pumping for 25 hours at a rate of 25 gpm

c. Pumping for 50 hours at a rate of 50 gpm

d. Pumping for 25 hours at a rate of 15 gpm.

4.6.5.4 Closed discharge. The pump shall be operated at maximum operating speed for 5 minutes at an ambient temperature of approximately 70°F against a totally closed discharge. No damage to the pump shall result.

4.6.5.5 Dry operation. The pump shall be operated at governed speed for a period of 1 hour without liquid. During this test, there shall be no detrimental effects on the pumping assembly.

* 4.6.6 Acoustic noise level test. The acoustic noise levels produced by the pumping assembly shall not exceed the values specified in 3.9 in any octave band at any of the measurement positions specified under 4.6.6.1. All acoustic measurements shall be made during rated capacity operation of the pumping assembly. Ambient background noise levels at all measurement positions shall be at least 10 db below the octave band sound pressure levels produced by the pumping assembly. The acoustic measurement instrumentation shall be as specified in 4.6.6.2.

* 4.6.6.1 Measurement positions. Measurements shall be made around the perimeter of the pumping assembly 5 feet above the ground level and 6 feet out from the edge of the assembly, facing towards the assembly. Three such measurements shall be made at equally spaced distances along both sides of the assembly, and two such measurements shall be made at the end of the assembly. Measurements shall be made at the operator's position which is directly in front of the control box, 5 feet above the ground level.

* 4.6.6.2 Instrumentation. The basic instrumentation shall consist of a sound level meter in conjunction with an octave band analyzer. This instrumentation shall meet the requirements specified in USA Standards S1.4 and Z24.10, except the octave band analyzer shall employ the preferred frequencies specified in USA Standard S1.6. An up-to-date calibration and frequency response of the entire measurement system, including microphone, shall be available to permit correction of the measured data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the

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response stated within ±2 db. An acoustical calibrator shall be used for an end-to-end system calibration from the microphone to the meter. The sound pressure level (in db ref 0.0002 dynes/CM^2 produced by the calibrator when used in conjunction with the particular microphone in the system shall be known within ±0.5 db.

4.6.7 Electromagnetic interference tests. The pumping assembly shall be tested for electromagnetic interference in accordance with method 7001 of MIL-STD-826.

4.6.8 Filter-separator test. The filter-separator shall be tested at 50 gpm in accordance with MIL-F-8901 with the following exception:

* 4.6.8.1 Inhibited fuel test. During the inhibited fuel test, a recirculating-type system will be permitted.

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* 4.6.9 Environmental tests. The following environmental tests shall be conducted on the pumping assembly in accordance with the specified methods of MIL-STD-810. Unless otherwise specified (see 6.2), an engineering evaluation may be performed in lieu of conducting the tests specified in 4.6.9.2 through 4.6.9.5 and 4.6.11. The engineering evaluation shall consist of a written detailed analysis of the pumping assembly and all its parts and components to satisfactorily accomplish the required environmental conditions.

4.6.9.1 Temperature tests

4.6.9.1.1 High temperature exposure and operation. The pumping assembly shall be installed in a test chamber and the high temperature exposure test conducted in accordance with procedure I of method 501. With the chamber temperature reduced to 120°F, the pump shall be started to insure satisfactory operation at this temperature. All controls, levers, and miscellaneous equipment shall be operated. The pump assembly shall be inspected to insure that plastic or other susceptible parts do not warp, twist, or change shape. The pump shall demonstrate the capability of pumping for 1 hour at rated capacity. The supply of fuel may remain outside the chamber for these tests. The engine oil temperature shall be recorded. There shall be no leakage, damage, or evidence of thermal degradation as a result of this test.

4.6.9.1.2 Low temperature exposure and operation. The pumping assembly and a supply of grade JP-4 fuel conforming to MIL-T-5624 shall be installed in a test chamber and a low temperature test conducted in accordance with procedure I of method 502. With the fuel temperature between -50° to -65°F and the pumping assembly

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at -65°F, the pump shall be started within 30 minutes, and the equipment operated to insure satisfactory performance at -65°F. All controls, levers, and miscellaneous equipment shall be operated a sufficient number of times to insure consistency of operation. A minimum of 500 gallons of fuel shall be pumped through the assembly. There shall be no leakage, damage, or evidence of thermal degradation as a result of this test.

4.6.9.1.3 Low temperature start at -20°F. Upon completion of the test specified in 4.6.9.1.2, the pumping assembly shall be cold-soaked for 24 hours at -20°F. Following this exposure, the pumping assembly shall be started within 10 minutes and operated for 10 minutes at -20°F without preheating of the engine or pumping assembly.

* 4.6.9.2 Humidity test. The humidity test shall be conducted in accordance with procedure I of method 507. There shall be no excessive corrosion resulting from this test.

* 4.6.9.3 Fungus test. The fungus test shall be conducted in accordance with procedure I of method 508. After exposure, the pumping assembly shall exhibit no effects of fungus attack of a deleterious nature.

* 4.6.9.4 Dust test. The dust test shall be in accordance with procedure I of method 510. No entrance of dust shall be permitted.

* 4.6.9.5 Salt-fog test. The salt-fog test shall be in accordance with procedure I of method 509. The pumping assembly shall exhibit no effects of a deleterious nature.

4.6.10 Mobility tests. Mobility tests shall be conducted in accordance with the applicable portions of MIL-M-8090 with the following exception and additions:

		Speed (MPH)		Distance
		Average	Maximum	(Miles)
a.	Level paved highway	40	45	1,000
b.	Graded gravel road	25	30	200
c.	Belgian-block course	15	18	200
d.	Cross-country terrain	15	18	100.

4.6.10.1 Post mobility test. Following the mobility tests the equipment shall be subjected to the tests specified in 4.6.2 and 4.6.3.

* 4.6.11 Air transportability test. The pumping assembly shall satisfactorily complete the air transportability tests of MIL-A-8421 for the Type C-123 aircraft.

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4.6.12 Reliability demonstration. Satisfactory reliability shall be considered demonstrated when both the following apply:

a. Prediction. A reliability prediction, using data, estimates, modifying factors, and procedures acceptable to the procuring activity shall yield prediction not less than the requirement of 3.5.1.

b. All tests specified herein shall be completed satisfactorily with no failure that could prevent safe or satisfactory performance of primary functions. Any test resulting in a failure shall be documented. Apparent failures that are caused by some unintended factor external to the pumping assembly (for example, test equipment failure) need not be counted as reliability failures if the causes can be substantiated to procuring activity satisfaction. Acceptance for reliability purposes shall not be made until, to the satisfaction of the procuring activity, each failure has been analyzed, the cause has been determined, design or other modifications have been made to preclude repetition of the failure or of related failures, the contractor has conducted successful retest of the failed equipment with the modifications included, and the contractor has taken action to reflect the modifications in equipment to be delivered. Any retest shall include all tests, in addition to a particular failed test, that could be influenced by the modifications.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-E-11275.

6. NOTES

6.1 Intended use. The PMU-27/M pumping assembly is intended for use in dispensing aircraft fuel from drums to all types of aircraft by the over-the-wing method.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. If hose sections are different than specified (see 3.5.9.3.1)

c. That MIL-M-008090E(USAF), dated 18 April 1966, or later issue should be used in conjunction with this specification (see 3.5.10)

d. If winterization kits are to be provided (see 3.7.7)

e. If test fuel is to be other than grade JP-5 fuel conforming to MIL-T-5624 (see 4.3.2)

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f. Location and conditions for preproduction testing (see 4.4)

g. When an engineering evaluation will not be acceptable in lieu of the testing specified in 4.6.9.2 through 4.6.9.5 and 4.6.11

h. Level of preservation, packaging, and packing required (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Failure. For reliability purposes, a failure is defined as any failure or malfunction that could prevent safe operation or that could prevent satisfactory operation in the primary functions of being towed to and from a using site, and of pumping fuel upon demand. Malfunctions that would not prevent such performance, such as loose bolts in noncritical applications, and preventive or scheduled maintenance done to preclude wear-out of time dependent failures shall not be considered reliability failures.

* 6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

g	gravitational units
gpm	gallons per minute
Hz	Hertz (cycles per second)
ID	inside diameter
mph	miles per hour
NPT	National taper pipe (thread)
psi	pounds per square inch
rpm	revolutions per minute

* 6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 84

Preparing activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on Reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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